

Constraints Essay: Calorie-Based Meal Finder

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Economic

Our project is an app or web app that allows users to log in and find fast-food meals and grocery options that fit their calorie goals.

The economic constraint is the cost of obtaining and maintaining the food and calorie data needed for both types of searches.

To control costs, our initial design should prioritize free development tools and publicly available nutrition information.

Using free data instead of a paid nutrition service could reduce expenses, but it could also limit food coverage and require more manual updates.

Security

Security factors into our project because users will log in and may associate personal calorie goals with their accounts.

The design should collect only the information necessary for account access and meal matching, avoiding unnecessary details such as medical history.

Account information and any saved goals should be protected through secure authentication and access controls that prevent users from viewing another person's information.

Using an established authentication service could simplify account protection, but its potential cost creates a trade-off between security support and affordability.

Social

Our intended users are people who want to identify fast-food meals and grocery options that fit a calorie target.

Supporting both food sources means the design should make clear whether a result represents a complete meal or an individual grocery item and identify the serving size used for its calorie count.

Because matching a calorie target does not establish a meal's overall nutritional quality, the app should present results as calorie matches without labeling them automatically healthy.